

Exercise for Sect. 2.5

Exercise

2.5.1

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if, for any $f, g \in L^2_\mu$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n f, g \rangle = \langle f, 1 \rangle \cdot \langle 1, g \rangle.$$

Proof: Suppose T is ergodic, by the continuity of inner product,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n f, g \rangle = \langle \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} U_T^n f, g \rangle.$$

By the mean ergodic theorem,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} U_T^n f = P_T f,$$

where P_T is the orthogonal projection to the T -invariant subspace. By the ergodicity of T , $P_T f$ is equal to constant c almost everywhere. Since T is measure-preserving,

$$\begin{aligned} \int_X f \cdot T^n d\mu &= \int_X f d\mu \\ \Rightarrow c &= \langle P_T f, 1 \rangle = \int_X \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} U_T^n f d\mu = \int_X f d\mu = \langle f, 1 \rangle. \end{aligned}$$

Thus,

$$\langle \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} U_T^n f, g \rangle = \langle P_T f, g \rangle = \langle c \cdot 1, g \rangle = c \cdot \langle 1, g \rangle = \langle f, 1 \rangle \cdot \langle 1, g \rangle.$$

Now, suppose for any $f, g \in L^2$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n f, g \rangle = \langle f, 1 \rangle \cdot \langle 1, g \rangle.$$

Suppose A is a T -invariant set,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n \chi_A, \chi_A \rangle = \langle \chi_A, 1 \rangle \cdot \langle 1, \chi_A \rangle = \mu(A)^2.$$

Since A is invariant, the left hand side can be written as $\int \chi_A \chi_A \, d\mu = \mu(A)$. Therefore,
 $\mu(A) = \mu(A)^2 \Rightarrow \mu(A) \in \{0, 1\}$ and T is ergodic. □

Exercise

2.5.2

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. For any function f in L^p_μ , $1 \leq p < \infty$, prove that

$$\frac{1}{n} \sum_{i=0}^{n-1} f(T^i x) \xrightarrow{L^p_\mu} f^*,$$

with $f^* \in L^p_\mu$ a T -invariant function.

Proof: Let $U_T : L^p_\mu \rightarrow L^p_\mu$ be the Koopman operator defined by $U_T f = f \circ T$. By the measure-preserving property of T , U_T is isometry. Then define $A_N f = \frac{1}{N} \sum_{n=0}^{N-1} U_T^n f$ be the first n average. Since U_T is isometry, $\|A_N\| \leq 1$.

For $f \in L^p_\mu$ and any $\varepsilon > 0$, we have a function $g \in L^\infty_\mu$ that $\|f - g\|_p \leq \frac{\varepsilon}{3}$. Since $\mu(X) = 1$, $g \in L^2_\mu$ and thus $A_N g \rightarrow g'$ by the mean ergodic theorem. Then, need to show $A_N g$ is Cauchy in L^p_μ .

For $p \in [1, 2)$, $\|\cdot\|_p \leq \|\cdot\|_2$ (in probability space) implies $A_N g$ is Cauchy in L^p_μ by $A_N g$ is Cauchy in L^2_μ . For $p \in (2, \infty)$, $\|A_N g\|_\infty \leq \|g\|_\infty$ and thus $|A_n g - A_m g| \leq 2\|g\|_\infty$ almost everywhere. Therefore,

$$\int |A_n g - A_m g|^p d\mu = \int |A_n g - A_m g|^{p-2} |A_n g - A_m g|^2 d\mu \leq (2\|g\|_\infty)^{p-2} \int |A_n g - A_m g|^2 d\mu.$$

Then $\|A_n g - A_m g\|_p \leq \left((2\|g\|_\infty)^{p-2} \|A_n g - A_m g\|_2^2 \right)^{\frac{1}{p}} \rightarrow 0$ as $n, m \rightarrow \infty$ by $A_n g$ is Cauchy in L^2_μ .

Thus, choosing N such that for all $n, m \geq N$, $\|A_n g - A_m g\|_p \leq \frac{\varepsilon}{3}$ and get

$$\|A_n f - A_m f\| \leq \|A_n(f - g)\|_p + \|A_n g - A_m g\|_p + \|A_m(f - g)\|_p \leq \varepsilon$$

By the completeness for L^p_μ , we have $A_n f \rightarrow f^*$.

$$\text{Since } U_T(A_n f) = A_n f + \frac{1}{n}(U_T^n f - f), \|U_T(A_n f) - A_n f\|_p \leq \frac{1}{n}(\|U_T^n f\|_p + \|f\|_p) = \frac{2\|f\|_p}{n}.$$

Thus, $U_T(A_n f) \rightarrow f^*$ and we also have $U_T(A_n f) \rightarrow U_T f^*$ by continuity of U_T .

Therefore, by uniqueness of limit in L^p_μ , f^* is T -invariant. $\square$

Exercise

2.5.3

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if

$A_N(f) \rightarrow \int f d\mu$ as $N \rightarrow \infty$ for all f in a dense subset of L^1_μ .

Proof: If T is ergodic, then just use the ergodic theorem.

Suppose $A_N(f) \rightarrow \int f d\mu$ as $N \rightarrow \infty$ for all f in a dense subset of L^1_μ . Let B be a T -invariant set in $\mathcal{B}$ and thus $\exists f_n \in L^1_\mu$ such that each f_n satisfies $A_N(f_n) \rightarrow \int f_n d\mu$ and $f_n \rightarrow \chi_B$ in L^1_μ .

Since $f_n \rightarrow \chi_B$ in L^1_μ , $\int f_n d\mu \rightarrow \mu(B)$. For any $\varepsilon > 0$, we have an $N_1 \in \mathbb{N}$ such that $\|f_n - \chi_B\|_1 < \varepsilon$ and $|\mu(B) - \int f_n d\mu| < \varepsilon$ for all $n > N_1$. Then, for all $N \in \mathbb{N}$ and fixed an $n > N_1$,

$$\|A_N(f_n) - \chi_B\|_1 = \left\| \frac{1}{N} \sum_{i=0}^{N-1} (f_n \circ T^i - \chi_B) \right\|_1 = \frac{1}{N} \left\| \sum_{i=0}^{N-1} (f_n - \chi_B) \circ T^i \right\|_1 \leq \frac{1}{N} (N \cdot \varepsilon) = \varepsilon.$$

Thus, for N is large such that $\|A_N(f_n) - \int f_n d\mu\|_1 < \varepsilon$,

$$\left\| \chi_B - \int \chi_B d\mu \right\|_1 \leq \|\chi_B - A_N(f_n)\|_1 + \|A_N(f_n) - \int f_n d\mu\|_1 + \left\| \int f_n d\mu - \int \chi_B d\mu \right\|_1 < 3\varepsilon.$$

Since χ_B take values on $\{0, 1\}$, we have $\mu(B) = \int \chi_B d\mu \in \{0, 1\}$ and T is ergodic. $\square$

Exercise

2.5.4

Show that

$$\lim_{N-M \rightarrow \infty} \frac{1}{N-M} \sum_{n=M}^{N-1} U_T^n f \rightarrow P_T f.$$

Proof: Since U_T is unitary and so continuous,

$$\lim_{N-M \rightarrow \infty} \frac{1}{N-M} \sum_{n=M}^{N-1} U_T^n f \stackrel{k=N-M}{=} U_T^M \left(\lim_{k \rightarrow \infty} \frac{1}{k} \sum_{n=0}^{k-1} U_T^n f \right) \rightarrow U_T^M (P_T f).$$

Since $P_T f \in I$ which is invariant subspace, $U_T^M P_T f = P_T f$.

□

Exercise

2.5.5

For any set B of positive measure in a measure-preserving system $(X, \mathcal{B}, \mu, T)$,

$$E = \{n \in \mathbb{N} : \mu(B \cap T^{-n}B) > 0\}$$

is syndetic: that is, there are finitely many integers $k_1, \dots, k_s$ with the property

that $\mathbb{N} \subset \bigcup_{i=1}^s E - k_i$.

Proof: Suppose B is a positive measure set,

$$\mu(B \cap T^{-n}B) = \int \chi_B \cdot T^n \cdot \chi_B \, d\mu = \langle U_T^n \chi_B, \chi_B \rangle.$$

By 2.5.4, we have

$$\lim_{N-M \rightarrow \infty} \frac{1}{N-M} \sum_{n=M}^{N-1} \langle U_T^n \chi_B, \chi_B \rangle = \langle P_T \chi_B, \chi_B \rangle.$$

Since P_T is orthogonal projection, $\langle P_T \chi_B, \chi_B \rangle = \|P_T \chi_B\|_2^2$. Also, since P_T is orthogonal projection and thus self-adjoint, $\langle P_T \chi_B, \chi_B \rangle = \langle \chi_B, P_T \chi_B \rangle = \langle \chi_B, \chi_B \rangle = \mu(B)$ since χ_B is T -invariant. Then, by Cauchy inequality, $0 < \mu(B)^2 = (\langle P_T \chi_B, \chi_B \rangle)^2 \leq \|P_T \chi_B\|_2^2 \cdot \|\chi_B\|_2^2 = \|P_T \chi_B\|_2^2$.

For all $\varepsilon > 0$, there is an $L \in \mathbb{N}$ such that for all M , $\left| \frac{1}{N-M} \sum \langle U_T^n \chi_B, \chi_B \rangle - \langle P_T \chi_B, \chi_B \rangle \right| < \varepsilon$ for all $N \geq L$. Take the $\varepsilon = \frac{\mu(B)}{2}$, we have $\frac{1}{L-M} \sum \langle U_T^n \chi_B, \chi_B \rangle > \frac{\mu(B)}{2} > 0$ for any M . Since each term is non-negative, the sum is greater than zero if there is some terms is positive. That is, we have no consecutive L intergers from any start point M is not in E . Thus, take $s = L$ and $k_i = i$ and proved E is syndetic. $\square$

Exercise

2.5.6

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. We say that T is totally ergodic if T^n is ergodic for all $n \geq 1$. Given $K \geq 1$ define a space $X^{(K)} = X \times \{1, \dots, K\}$ with measure $\mu^{(K)} = \mu \times \nu$ defined on the product σ -algebra $\mathcal{B}^{(K)}$, where $\nu(A) = \frac{1}{K} |A|$ is the normalized counting measure defined on any subset $A \subseteq \{1, \dots, K\}$, and a $\mu^{(K)}$ -preserving transformation $T^{(K)}$ by

$$T^{(K)}(x, i) = \begin{cases} (x, i+1) & \text{if } 1 \leq i < K \\ (Tx, 1) & \text{if } i = K \end{cases}$$

for all $x \in X$. Show that $T^{(K)}$ is ergodic with respect to $\mu^{(K)}$ if and only if T is ergodic with respect to μ , and that $T^{(K)}$ is not totally ergodic if $K > 1$.

Proof: If T is ergodic w.r.t. μ , let $B \in \mathcal{B}^{(K)}$ be an $T^{(K)}$ -invariant set. By the invariance, we have if $(x, k) \in B$ for some k , $(x, k) \in B$ for all k and $B|_X$ (B restrict on X) should be T -invariant set. Thus, $\mu^K(B) = \mu(B|_X) \in \{0, 1\}$ implies the ergodicity of $T^{(K)}$.

If $T^{(K)}$ is ergodic w.r.t. $\mu^{(K)}$, and $B \in \mathcal{B}$ is T -invariant. Then $B^{(K)} = \{(x, k) : x \in B, k \in \{1, \dots, k\}\}$ is $T^{(K)}$ -invariant with measure $\mu^{(K)}(B^{(K)}) = \mu(B) \in \{0, 1\}$. Therefore, T is ergodic.

For $K > 1$, $(T^{(K)})^K$ is not ergodic with the invariant set $\{(x, 1) : x \in B\}$ for some invariant set B of T and thus $T^{(K)}$ is not totally ergodic. □